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Detailed mark scheme

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AQA A Level Further Mathematics (7367)

Topic C: Matrices

Sub topic C6: Singular Matrices (Properties of Singular Matrices)

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic	**C: Matrices**
Sub topic	**C6: Singular Matrices (Properties of Singular Matrices)**

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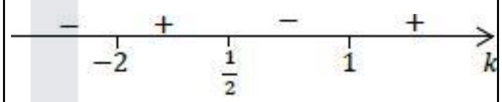
To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Substitutes $k = 1$ and correctly calculates the determinant, and concludes that the matrix is singular.	AO2.2a	B1	determinant = $4 \times 1 - 2 \times 2 = 0$ ∴ the matrix is singular **AG**
(b)	Finds determinant in terms of k. Allow one error.	AO1.1a	M1	determinant = $k(5 - k) - 2(k^3 + 1)$
	Obtains a correct inequality in k.	AO1.1b	A1	$5k - k^2 - 2k^3 - 2 < 0$
	Obtains three correct critical values.	AO1.1b	A1	$2k^3 + k^2 - 5k + 2 > 0$ $(k - 1)(2k^2 + 3k - 2) > 0$ $(k - 1)(2k - 1)(k + 2) > 0$ 
	Deduces one correct region. FT their three real distinct critical values if given as $a < k < b$, $k > c$ (o.e.) where $a < b < c$	AO2.2a	A1F	
	Deduces the other correct region. FT their three real distinct critical values if given as $a < k < b$, $k > c$ (o.e.) where $a < b < c$ Condone the use of 'and'.	AO2.2a	A1F	$-2 < k < \frac{1}{2}$, $k > 1$
				Total 6 marks

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Q2.

(a) $$\begin{vmatrix} -2 & 1 & 2k \\ -1 & 1 & k+1 \\ 2 & k-1 & 1 \end{vmatrix} = -2\begin{vmatrix} 1 & k+1 \\ k-1 & 1 \end{vmatrix} + 1\begin{vmatrix} 1 & 2k \\ k-1 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 2k \\ 1 & k+1 \end{vmatrix}$$

correctly expanding by any row or column

M1

$= -2[1 - (k + 1)(k - 1)] + [1 - 2k(k - 1)] + 2[k + 1 - 2k]$

correct unsimplified expansion of 2 × 2 determinants

A1

$= -2[1 - k^2 + 1] + [1 - 2k^2 + 2k] + 2[1 - k]$

$= -4 + \cancel{2k^2} + 1 - \cancel{2k^2} + \cancel{2k} + 2 - \cancel{2k}$

$= -1$

−1 obtained

A1cso

either all k's cancel *or* independent of k *etc*

comment required (must score previous 3 marks

E1

4

Alternative:

$$\begin{vmatrix} -2 & 1 & 2k \\ -1 & 1 & k+1 \\ 2 & k-1 & 1 \end{vmatrix} \quad \begin{array}{l} r_1 \to r_1 - 2r_2 \\ r_3 \to r_3 + 2r_2 \end{array}$$

$$= \begin{vmatrix} 0 & -1 & -2 \\ -1 & 1 & k+1 \\ 0 & k+1 & 2k+3 \end{vmatrix} \quad r_3 \to r_3 + (k+1)r_1$$

correctly expanding by any row or column after row operations

(M1)

$$= \begin{vmatrix} 0 & -1 & -2 \\ -1 & 1 & k+1 \\ 0 & 0 & 1 \end{vmatrix} = -1$$

correct expansion unsimplified

(A1)

−1 obtained

(A1)

either all k's cancel *or* independent of k etc

comment required

(E1)

(4)

(b) (i) Identifying that $k = 3$

$k = 3$

B1

Value of determinant ≠ 0 (or = −1 etc) therefore equations are consistent

ft answer (a) if 0 ⇒ inconsistent

E1F

2

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(ii) 3 planes intersect in a unique point

B1

1

[7]

Q3.

(a) $\begin{vmatrix} 2 & 1 & 3 \\ 5 & -2 & a+1 \\ a & 2 & 4 \end{vmatrix} = a^2 + 3a - 10$

Attempt at det of coeff matrix

M1

Correct (accept unsimplified)

A1

Equating to 0 and solving quadratic in a

SC: B1 for verifying $a = 2$

m1

$a = 2, -5$

B1 for verifying $a = -5$

A1

4

(b) $\begin{aligned} 2x + y + 3z &= 3 \\ 5x - 2y + 3z &= 3 \\ 2x + 2y + 4z &= b \end{aligned}$

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B1

Eliminations leading to two equations in two variables

M1

Further elimination leading to value of b

m1

$b = 4$

A1

ALT: Finding two variables in terms of third

eg $y = x$ *and* $z = 1 - x$

(M1)

Substituting into third equation

(m1)

$b = 4$

(A1)

4

[8]

Q4.

(a) $\begin{vmatrix} 1 & -2 & k \\ k+1 & 3 & 0 \\ 2 & 1 & k-1 \end{vmatrix} = 3k^2 - 2k - 5$

Attempt at det. of coefft. mtx.

M1

Correct

A1

Setting det. = 0 and solving for k

M1

$k = \frac{5}{3}, -1$

A1

4

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(b) $5 = x - 2y + \frac{5}{3}z$

$k = \frac{5}{3} \Rightarrow \frac{5}{3} = \frac{8}{3}x + 3y$

ft

B1

$3 = 2x + y + \frac{2}{3}z$

Eliminating one variable twice

M1

$8x + 9y = 5/15y - 8z = -21 \,/\, 5x + 3z = 11$

Correct eqn. once; twice

A1;A1

$5 = x - 2y - z$
$k = -1 \Rightarrow -1 = 3y$
ft

B1

$3 = 2x + y - 2z$
Eliminating one variable twice

M1

$x - z = \frac{13}{3} / 2x - 2z = \frac{10}{3} / 5x - 5z = 11$
Inconsistency correctly shown

A1

OR
$y = -\frac{1}{3}$ and $y = -\frac{7}{5}$ found

A1 **8**

[12]

Q5.

(a) $\begin{bmatrix} 1 & 4 & 2 \\ -1 & 2 & 6 \end{bmatrix}\begin{bmatrix} k & 1 \\ 2 & -1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} k+14 & -1 \\ 22-k & 3 \end{bmatrix}$

***PQ** a 2 × 2 matrix*

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M1

At least one element in C_1 correct

A1

All correct

A1 **3**

(b) Det(**PQ**) = $3k + 42 + 22 - k$
Det of a square matrix attempted and equated to zero

M1

$= 2k + 64 = 0$
$k = -32$
ft in 2 × 2 case only (linear eqns.)

A1 **2**

[5]

Q6.

(a) (i) **AB** = a 3 × 3 matrix

M1

$$= \begin{pmatrix} 3 & 2 & t+1 \\ 1 & 2 & t-1 \\ 3 & 2 & t+1 \end{pmatrix}$$

At least 5 elements correct, incl. at least one from C_3

A1

All elements correct

A1

3

(ii) **BA** = a 2 × 2 matrix

M1

$$= \begin{pmatrix} 2 & 2 \\ t & t+4 \end{pmatrix}$$

A1

2

(b) $R_1 = R_2$ ($\Rightarrow$ det **AB** = 0)

Or expanding and ***showing*** *det = 0*

B1

1

(c) **BA** = $\begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix} = 2\sqrt{2}\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

M1 A1

E: enlargement s.f. $2\sqrt{2}$

B1

F: Rotation

NB: Rotation bit may be sorted completely separately in which case marks are split 3 + 3

M1

clockwise (about O) thro’ 45°

Or − 45°, 315°

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A1 A1

6

[12]

Q7.

(a) Setting det **P** $(2t - 6 + 2 - 3t + 8 - 1 = 3 - t) = 0$

M1

$\Rightarrow t = 3$

A1

2

(b) (i) $\begin{bmatrix} 6 & 0 & 0 \\ -7t-21 & 3-t & 15+5t \\ 0 & 0 & 6 \end{bmatrix}$

Multiplication attempt with ≥ 3 correct elements

M1

Any one row or column

A1

All correct

A1

3

(ii) When $t = -3$, **PQ** = $6\mathbf{I}$

The 6 may be implicit

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B1 B1

2

(iii) $\mathbf{Q}^{-1} = \frac{1}{6}\mathbf{P} = \frac{1}{6}\begin{bmatrix} 2 & 1 & 1 \\ 1 & -3 & -2 \\ 3 & 2 & 1 \end{bmatrix}$

Allow for $\frac{1}{6}$ ***P*** *or* $\frac{1}{36}$ *(6**P**)*

B1

1

(c) Enlargement (centre O) sf 6

M1 A1

2

[10]

Examiner reports

Q1.

The majority of students were able to calculate the determinant and show that it was equal to zero. However, a significant minority failed to make a conclusion after a correct calculation.

In part (b), most students were able to set up a correct inequality, although some wasted time searching for a root, having missed the connection with part (a). Those students who found the correct critical values, generally went on to identify the correct regions.

Q2.

Expansions of determinants are well understood, although there were often sign errors with the terms 1, –1, $k + 1$ and $k – 1$. Several students failed to comment on their result to show they had understood what was meant by 'independent of k'. Many students failed to see the link between the determinant in part (a) and the rest of the question, with some starting again at each part. Furthermore, it is clear that students do not understand the concept of consistency of a system of equations. Many responses were seen that stated 'consistent' and gave a prism as the configuration or 'inconsistent' and sheaf.

Q3.

In part (a), most candidates realised that the determinant of the coefficients on the left-hand sides of the equations must be equated to zero, and thus obtained the required values efficiently. Solutions to part (b) were often marred by a failure to indicate which equations were being used at each step. Some candidates used the slightly dubious technique of replacing one of the variables by zero, while others produced a whole page of work and still failed to reach a value for b.

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Q4.

This question was surprisingly well attempted, as the algebra that goes with the systems of equations work is usually found tough enough to guarantee lots of mistakes. Almost all candidates found the determinant of the coefficient matrix in part (a) and solved the resulting quadratic equation with ease. Other approaches generally got nowhere. The work in part (b) was also competently attempted by the majority, apart from silly arithmetical slips which again probably would have been noticed had a very quick check been made.

Q5.

This was a straightforward starter to the paper, and was generally found to be so by candidates.
Almost all candidates knew what to do, in principle, although there were unexpectedly large numbers of arithmetical (particularly sign) errors in part (b) when finding the required value of k.

Q6.

The two product matrices **AB** and **BA** in part (a) of this question were very popular and the majority of candidates gained all 5 marks. The rest of the question really was poorly done on the whole, but this was the most demanding work on the paper. In part (b), many of candidates noted that det (**AB**) was zero, but failed to explain why. In part (c), most seemed to resort to guesswork, especially regarding the transformation F. A shear was, marginally, the most popular choice, with a 90^{o} rotation close behind. The majority of candidates seemed to have no method of approach to the problem. Those that took a scalar factor out of the matrix usually contented themselves with 2, 4 or 8 as the scale factor of the enlargement E; thereby leaving themselves with a matrix they simply did not know what to do with. Those who chose "rotation" as their answer then did so from a matrix with det ≠ 1 and consequently did not gain all the marks available.

Q7.

This proved to be a profitable question for many candidates. In particular, the multiplication of two 3x3 matrices was handled very competently indeed, even though algebraic terms were involved. The odd error arising here did not prevent candidates from doing good work in the later parts of the question. They were not required to check that all entries of the product matrix gave consistent values for both t and k.

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